

2026 Chantilly Math Competition

Middle School Division - Individual Round Solutions

1. (9 pts) Given that x is an integer satisfying

$$x^x = 2^{2048},$$

compute x .

Victor Moldoveanu

Solution If some prime $p \neq 2$ divides x , then it also divides $x^x = 2^{2048}$, which is absurd.

Therefore, $x = 2^y$ for some integer y , so $2^{y \cdot 2^y} = 2^{2048}$, so $y \cdot 2^y = 2048 = 2^{11}$.

If some prime $p \neq 2$ divides y , then it also divides $y \cdot 2^y = 2^{11}$, which is absurd.

Therefore, $y = 2^z$ for some integer z , so $2^z \cdot 2^{2^z} = 2^{z+2^z} = 2^{11}$, so $z + 2^z = 11$.

This is true at $z = 3$, and the LHS is clearly increasing w.r.t z , so that's the only solution.

Therefore, $x = 2^y = 2^{2^z} = 2^{2^3} = 2^8 = \boxed{256}$.

2. (9 pts) If a and b are integers such that $a - b = 7$ and $130 \leq a^2 - b^2 \leq 139$, what is $a^3 - b^3$?

Ryan Chan

Solution $7 = (a - b) \mid (a - b)(a + b) = a^2 - b^2$ so $a^2 - b^2 = 133$ and $a + b = 19$.

Therefore, $a = 13$ and $b = 6$, so $a^3 - b^3 = 2197 - 216 = \boxed{1981}$.

3. (10 pts) The positive integers from 1 to 7 inclusive are written on a blackboard. While there are more than 2 numbers on the blackboard, Simon will choose a pair of numbers from the blackboard uniformly at random, and erase the smaller of the numbers. Afterwards, there are 2 numbers on the blackboard, and the probability that they are 6 and 7 can be expressed as $\frac{a}{b}$ for relatively prime positive integers a and b . Compute $a + 10b$.

Aryan Raj

Solution We just need the pair of 6 and 7 to never be chosen.

Therefore, the desired probability is

$$\left(1 - \frac{1}{\binom{7}{2}}\right)\left(1 - \frac{1}{\binom{6}{2}}\right)\left(1 - \frac{1}{\binom{5}{2}}\right)\left(1 - \frac{1}{\binom{4}{2}}\right)\left(1 - \frac{1}{\binom{3}{2}}\right) = \frac{4}{9}.$$

Therefore, the answer is $4 + 10(9) = \boxed{94}$.

4. (10 pts) Let $f(x) = x^2 + ax + b$ and $g(x) = x^2 + cx + d$.

Given that $f(3) = f(5) = 0$ and $g(a) = g(b) = 0$, compute $|g(1)|$.

Arsenii Zharkov

Solution $f(x) = (x - 3)(x - 5) = x^2 - 8x + 15$ so $a = -8$ and $b = 15$.

$g(x) = (x - a)(x - b) = (x + 8)(x - 15)$ so $|g(1)| = |(1 + 8)(1 - 15)| = 9 \cdot 14 = \boxed{126}$.

5. (11 pts) Square $ABCD$ has side length 100. Point X lies on line segment AD . Compute the smallest possible value of AX such that the circle passing through A , X , and B intersects line segment CD .

Aryan Raj

Solution Let the answer be x .

Then, the radius of the circle is $100 - \frac{x}{2}$.

Therefore, $(100 - \frac{x}{2})^2 = 50^2 + (\frac{x}{2})^2$.

Therefore, $100(100 - x) = 50^2$, so $x = \boxed{75}$.

6. (11 pts) Let $f : \mathbb{Z} \rightarrow \mathbb{Z}$ be a function such that

$$f(x+1) + f(x+2) = f(x) + f(x+3)$$

for all integers x . Given $f(4) = 1$ and $f(6) = 7$, compute $f(14) + f(76)$.

Aryan Raj

Solution We know $f(x-2) + f(x+1) = f(x-1) + f(x)$ and $f(x-1) + f(x+2) = f(x) + f(x+1)$.

Adding both equations gives $f(x-2) + f(x-1) + f(x+1) + f(x+2) = f(x-1) + 2f(x) + f(x+1)$.

Subtracting $f(x-1) + f(x+1)$ from both sides gives $f(x-2) + f(x+2) = 2f(x)$.

This rearranges into $f(x+2) - f(x) = f(x) - f(x-2)$, so

$$f(4), f(6), f(8), \dots$$

forms an arithmetic sequence.

The first term is $f(4) = 1$, and the common difference is $f(6) - f(4) = 7 - 1 = 6$.

Therefore, $f(4 + 2k) = 1 + 6k$ for all integers k .

Therefore, $f(14) + f(76) = f(4 + 2(5)) + f(4 + 2(36)) = 1 + 6(5) + 1 + 6(36) = 2 + 6(41) = \boxed{248}$.

7. (12 pts) 41 students compete in CMC. They randomly form teams of 3 and 4 for the team round. Having more people on your team is better, so the number of teams of 4 is maximized. Alice and Bob are 2 of the 41 CMC competitors. Given that the probability of Alice and Bob being on the same team can be expressed as $\frac{a}{b}$ for relatively prime positive integers a and b , compute $a + b$.

Aryan Raj

Solution Clearly, there will be 8 teams of 4 and 3 teams of 3.

Therefore, the desired probability is

$$\frac{8\binom{4}{2} + 3\binom{3}{2}}{\binom{41}{2}} = \frac{57}{820}.$$

Therefore, the answer is $57 + 820 = \boxed{877}$.

8. (12 pts) Given that

$$\sqrt{x} + 1 = \sqrt{x + 2\sqrt{1 + \sqrt{\frac{2+x}{10}}} + 1}$$

and x can be expressed as $\frac{m}{n}$ for relatively prime positive integers m and n , compute $m + n$.

Krithik Manoharan

Solution Square both sides to get

$$x + 1 + 2\sqrt{x} = x + 1 + 2\sqrt{1 + \sqrt{\frac{2+x}{10}}},$$

so $x = 1 + \sqrt{\frac{2+x}{10}}$. Then, $x - 1 = \sqrt{\frac{2+x}{10}}$, so $x^2 - 2x + 1 = \frac{2+x}{10}$, or $10x^2 - 21x + 8 = 0$. This factors as $(5x - 8)(2x - 1) = 0$, so $x = \frac{8}{5}$ or $\frac{1}{2}$. However, after plugging these in to the equation, we find that $\frac{1}{2}$ doesn't work, giving $x = \frac{8}{5}$, for an answer of $8 + 5 = \boxed{13}$.

9. (13 pts) Tyrone draws a triangle. The number of distinct pairs of angles such that one angle is twice the other angle is maximized. Given that the sum of all distinct possible values of the smallest angle in the triangle is
- $\frac{m}{n}$
- degrees for relatively prime positive integers
- m
- and
- n
- , compute
- $m + n$
- .

Aryan Raj

Solution We claim there can't be 3 pairs. Let 2 angles be k and $2k$.

Then, the third angle must be $4k$ or k , neither of which are twice or half of k .

Therefore, the maximum is 2 pairs, and we will show that this is achievable.

By Pigeonhole Principle, these 2 pairs must share some common vertex.

Let the measure of the angle at that vertex be x .

Then, we have the following 3 cases.

$x + 2x + 2x = 180^\circ$ so $x = 36^\circ$ and the smallest angle is $x = 36^\circ$.

$x + \frac{x}{2} + \frac{x}{2} = 180^\circ$ so $x = 90^\circ$ and the smallest angle is $\frac{x}{2} = 45^\circ$.

$x + \frac{x}{2} + 2x = 180^\circ$ so $x = (\frac{360}{7})^\circ$ and the smallest angle is $\frac{x}{2} = (\frac{180}{7})^\circ$.

The total is $36^\circ + 45^\circ + (\frac{180}{7})^\circ = (\frac{747}{7})^\circ$.

Therefore, the answer is $747 + 7 = \boxed{754}$.

10. (13 pts) Given

$$1 \cdot 1! + 2 \cdot 2! + 3 \cdot 3! + 4 \cdot 4! + \cdots + 2025 \cdot 2025! = n! + m$$

for some positive integer n and real number m such that $|m| < n$, compute $m + n$.

Sujith Miriyala

Solution Each term in the sum is in the form $(x-1)(x-1)!$, where x is an integer that ranges from 2 to 2026 inclusive.

Expanding yields $x(x-1)! - (x-1)! = x! - (x-1)!$.

Thus, the sum reduces to $2! - 1! + 3! - 2! + 4! - 3! + 5! - 4! + \cdots + 2026! - 2025!$.

Notice that everything cancels except $2026!$ and $-1!$, yielding $2026! - 1$.

Therefore, $m + n = 2026 - 1 = \boxed{2025}$.

11. (14 pts) A positive integer n is nice if and only if there is some positive integer k such that k^n leaves a remainder of 6 when divided by 7. Compute the sum of all nice positive integers less than 67.

Aryan Raj

Solution We claim n is nice if and only if n is odd.

When n is odd, $n = 2m + 1$, so

$$6^n = 6^{2m+1} = 6 \cdot 36^m \equiv 6 \cdot 1^m = 6 \pmod{7}.$$

When n is even, k^n is some perfect square m^2 , and

$$m^2 \in \{1^2, 2^2, \dots, 6^2\} = \{1, 2, 4\} \pmod{7}.$$

Therefore, the answer is $1 + 3 + \dots + 65 = 33^2 = \boxed{1089}$.

12. (14 pts) Let $ABCD$ be a square with side length 8. Let E be a point on the circle passing through points A , B , C , and D . Let F be the midpoint of line segment EA . Let G be the midpoint of line segment FB . Given that the longest possible length of line segment GC can be expressed as $\sqrt{a} + \sqrt{b}$ where a and b are positive integers, compute ab .

Aryan Raj

Solution Let A , B , C , and D have coordinates $(4\sqrt{2}, 0)$, $(0, 4\sqrt{2})$, $(-4\sqrt{2}, 0)$, and $(0, -4\sqrt{2})$, respectively.

The locus of point E is the circle with center $(0, 0)$ and radius $4\sqrt{2}$.

By taking a homothety from A with ratio $\frac{1}{2}$, the locus of point F is the circle with center $(2\sqrt{2}, 0)$ and radius $2\sqrt{2}$.

By taking a homothety from B with ratio $\frac{1}{2}$, the locus of point G is the circle with center $(\sqrt{2}, 2\sqrt{2})$ and radius $\sqrt{2}$.

The distance between C , which is $(-4\sqrt{2}, 0)$, and $(\sqrt{2}, 2\sqrt{2})$, is $\sqrt{58}$, so the longest possible length of GC is $\sqrt{2} + \sqrt{58}$. Therefore, the answer is $2 \cdot 58 = \boxed{116}$.

13. (15 pts) Compute the remainder when 9^{67} is divided by 1000.

Aryan Raj

Solution

$$9^{67} = (10 - 1)^{67} \equiv -\binom{67}{2}10^2 + \binom{67}{1}10 - \binom{67}{0} \equiv -100 + 670 - 1 = \boxed{569} \pmod{1000}.$$

14. (15 pts) What real number a makes the equation $x^3 - 57x^2 + ax - 343 = 0$ have 3 distinct solutions for x such that the square of 1 of them equals the product of the other 2?

Achyut Dipukumar

Solution Let the 3 real solutions for x be p , q , and r .

WLOG, let $p^2 = qr$.

Equating coefficients in $(x - p)(x - q)(x - r) = x^3 - 57x^2 + ax - 343$ gives

$$p + q + r = 57, pq + qr + rp = a, pqr = 343.$$

Since $p^2 = qr$, we have $343 = pqr = p^3$, so $p = 7$, so $qr = 49$.

Since $p + q + r = 57$, we have $q + r = 50$.

Therefore, $a = pq + qr + rp = p(q + r) + qr = 7(50) + 49 = \boxed{399}$.

15. (16 pts) Triangles ABC and DEF both have positive integer area. Points D , E , and F lie on line segments AB , BC , and CA , respectively. Given that

$$\frac{AD}{DB} = \frac{BE}{EC} = \frac{CF}{FA} = \frac{6}{7},$$

compute the smallest possible area of triangle DEF .

Aryan Raj

Solution For brevity, let $[XYZ]$ denote the area of triangle XYZ .

$$[ADF] = \frac{AD}{AB} \cdot \frac{AF}{AC} \cdot [ABC] = \frac{6}{13} \cdot \frac{7}{13} \cdot [ABC] = \frac{42}{169}[ABC].$$

By symmetry,

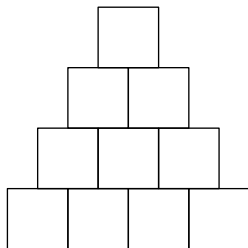
$$[ADF] = [BED] = [CFE] = \frac{42}{169}[ABC].$$

Therefore,

$$[DEF] = [ABC] - [ADF] - [BED] - [CFE] = \frac{169 - 3 \cdot 42}{169}[ABC] = \frac{43}{169}[ABC].$$

When $[ABC] = 169$, we get the smallest possible positive integer value of $[DEF] = \boxed{43}$.

16. (16 pts) The Chantilly Math Club is painting a mural. They need to paint each of the 10 squares in the picture below with one of four colors: black, gray, purple or white.



They must follow these rules:

1. No two squares that touch each other may have the same color.
2. A black square cannot touch a gray square (because black and gray look very similar)

In how many ways can the Chantilly Math Club paint the mural?

Ryan Chan

Solution Label the middle square of the second row as the *center square* (the square from $(1.5, 1)$ to $(2.5, 2)$). It touches 6 other squares: the two adjacent in its row, the two directly below (sharing half-edges), and the two directly above (sharing half-edges).

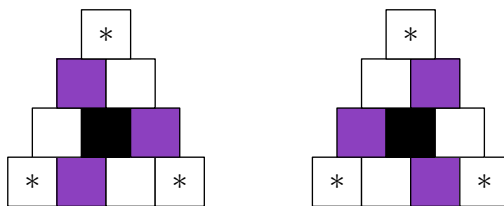
We casework on the color of the center square.

Case 1: The center square is black

If the center is black, then any square touching it cannot be black (Rule 1) and cannot be gray (Rule 2). So the 6 neighbors of the center must all be colored using only **purple** and **white**. Because adjacent neighbors may not share a color, these 6 squares force one of the **two** patterns below.

In each pattern, the three remaining squares (marked by *) each touch one purple and one white square, so the only available colors for each * square are **black or gray**. Thus there are 2^3 choices for the * squares, and

$$\#(\text{murals with black center}) = 2 \cdot 2^3 = 16.$$



Case 2: The center square is gray

This is identical to Case 1 (swap the words “black” and “gray”). So,

$$\#(\text{murals with gray center}) = 16.$$

Case 3: The center square is purple

If the center is purple, then its 6 neighboring squares cannot be purple. A convenient way to satisfy the restrictions is to force those 6 squares into two alternating types:

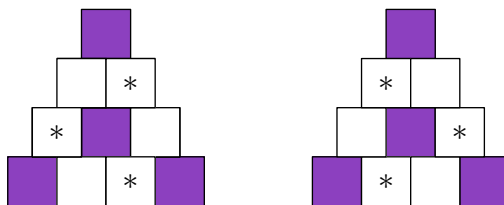
(white squares) and (* squares, each either black or gray).

There are exactly **two** such alternating arrangements (shown below).

In either arrangement, each corner square touches a white square and an * square. It cannot be white, and it cannot be black/gray because it would have to avoid both possibilities for the adjacent * square (also black cannot touch gray). Hence each corner square is forced to be **purple**.

Finally, the three * squares do not touch each other, so each can independently be black or gray, giving 2^3 choices. Thus,

$$\#(\text{murals with purple center}) = 2 \cdot 2^3 = 16.$$



Case 4: The center square is white

This is equivalent to Case 3 (swap the words “purple” and “white”), so

$$\#(\text{murals with white center}) = 16.$$

Total

Adding all four cases,

$$16 + 16 + 16 + 16 = 64.$$

Therefore, the mural can be painted in 64 ways.

- TB. (Tiebreak) Shubham loves chess, but only has the black pieces. Therefore, he has a chessboard, which is a rectangular grid with 8 squares on each side, and 16 black chess pieces. Assuming all black chess pieces are indistinguishable and each of the 64 squares on the chessboard has at most 1 piece, estimate the number of ways Shubham can place the 16 black chess pieces on the chessboard such that each of the 8 rows and each of the 8 columns has exactly 2 black pieces in it.

Aryan Raj

Solution We biject each way to a graph with 16 vertices $r_1, c_1, r_2, c_2, \dots, r_8$, and c_8 by drawing an edge between r_i and c_j iff there is a chess piece in the i th row and j th column.

Since each row and each column has 2 chess pieces in it, all 16 vertices have degree 2.

This means that the graph can be decomposed into cycles.

Since the graph is bipartite, each of these cycles must have even length.

Let there be k_l cycles of length $2l$, where $l \geq 2$.

Then, there are

$$\frac{(8!)^2}{\prod_{i=2}^{\infty} ((2i)^{k_i} k_i!)}$$

distinct graphs resulting from labelling the vertices.

Since $\sum_{i=2}^{\infty} 2ik_i = 16$, we have $\sum_{i=2}^{\infty} ik_i = 8$, so the only cases are:

- (a) $k_8 = 1$
- (b) $k_6 = 1, k_2 = 1$
- (c) $k_5 = 1, k_3 = 1$
- (d) $k_4 = 2$
- (e) $k_4 = 1, k_2 = 2$
- (f) $k_3 = 2, k_2 = 1$
- (g) $k_2 = 4$

Therefore, the number of ways is

$$\frac{(8!)^2}{16} + \frac{(8!)^2}{12 \cdot 4} + \frac{(8!)^2}{10 \cdot 6} + \frac{(8!)^2}{8^2 \cdot 2!} + \frac{(8!)^2}{8 \cdot 4^2 \cdot 2!} + \frac{(8!)^2}{6^2 \cdot 2! \cdot 4} + \frac{(8!)^2}{4^4 \cdot 4!} = \boxed{187530840}.$$